

A meta-analysis of maternal cured meat consumption during pregnancy and the risk of childhood brain tumors.

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OBJECTIVE: N-Nitroso compounds (NOCs) are recognized neural carcinogens in animal models and are suspected human carcinogens. A meta-analysis was performed examining the possible association of maternal intake of cured meat (an important source of dietary NOCs) during pregnancy and the risk of pediatric brain tumors. METHODS: Data from epidemiological studies were pooled using a general variance-based meta-analytic method employing confidence intervals described by Greenland in 1986. The outcome of interest was a summary relative risk (RR) reflecting the risk of childhood brain tumor (CBT) development associated with maternal intake of cured meats during pregnancy. Sensitivity analyses were performed when necessary to explain any observed statistical heterogeneity. RESULTS: Seven observational studies were found that met the protocol-specified inclusion criteria. Analysis for heterogeneity demonstrated a lack of statistical heterogeneity ($p = 0.59$), indicating that the data could be statistically combined. Pooling data from the 6 reports containing data on maternal cured meat intake of all types yielded an RR of 1.68 (1.30-2.17), being a statistically significant result. Analyzing CBT risk by type of cured meat ingested showed that hot dog consumption increased CBT risk by 33% (1.08-1.66), with a similar increase shown by frequent ingestion of sausage, i.e. 44%. CONCLUSION: The data provide support for the suspected causal association between ingestion of NOCs from cured meats during pregnancy and subsequent CBT in offspring. Limitations in study design preclude definitive conclusions, but the relationship warrants exploration via additional observational and laboratory-based studies.

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